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Power Plant Supply Chains

Assessing the Supply Chains of Critical Power Plant
Components and Capital Equipment

Nuclear Technologies and National Security Directorate

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Power Plant Supply Chains

Assessing the Supply Chains of Critical Power Plant
Components and Capital Equipment

by

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List of Acronyms

AMMTO	Advanced Materials and Manufacturing Technologies Office
ARPA-E	Advanced Research Projects Agency - Energy
BOP	Balance of Plant
CM	Critical Minerals
DCS	Distributed Control System
DFC	U.S. International Development Finance Corporation
DLA	Defense Logistics Agency
DOE	U.S. Department of Energy
DOW	U.S. Department of War
DPA	Defense Production Act
EIA	U.S. Energy Information Administration
ESR	Electroslag Remelting
EXIM	Export-Import Bank of the United States
FAST-41	Fixing America's Surface Transportation Act Title 41
GIS	Gas Insulated Switchgear
GOES	Grain Oriented Electrical Steel
GT	Gas Turbine
HRSG	Heat Recovery Steam Generator
I/O	Input/Output
MEA	Middle East and Africa
NGCC	Natural Gas Combined Cycle
NPI	Nickel Pig Iron
OEM	Original Equipment Manufacturer
PLC	Programmable Logic Controller
PRC	People's Republic of China
R&D	Research and Development
REE	Rare Earth Element
QA	Quality Assurance
ROW	Rest of World
SCADA	Supervisory Control and Data Acquisition
SC/DS	Single-Crystal/Directionally Solidified
ST	Steam Turbine
USGS	United States Geological Survey
VAR	Vacuum Arc Remelting

Executive Summary

Reliable deployment of power-generation infrastructure depends on complex global supply chains for materials, manufacturing, and specialized equipment. This study evaluates supply-chain exposure across major power-plant subsystems to identify potential bottlenecks that could affect near-term deployment timelines. The study covers power-generation supply chains broadly, using natural gas combined-cycle (NGCC) as the reference for the detailed subsystem assessment because natural gas accounts for the largest share of U.S. electricity generation and is estimated to remain a major generation source through 2030 (EIA, 2026). Many of the subsystems, materials, components, and manufacturing processes evaluated in this study are also relevant to coal, geothermal, hydropower, and other generation technologies and were captured in the broader system-level screening. The assessment approach is summarized in Table I-1.

Table I-1: System-Level Assessment

Assessment Step	Key Outcome
Full-system screening	11 power-plant subsystems evaluated
Risk indicators assessed	Foreign dependence, critical materials, domestic capacity, and manufacturing complexity
Subsystems prioritized	6 subsystems selected
Components identified	13 relevant components
Detailed assessments	5 components selected for deep-dive analysis

Supply-chain risk is present across all eleven power-plant subsystems evaluated but is concentrated in a limited number of material inputs and specialized manufacturing stages (See example in Table I-2). These include critical alloy inputs, specialized manufacturing processes, and electrical and control systems, where dependencies on constrained materials, limited domestic capacity, or specialized equipment create potential bottlenecks.

Table I-2: Cross-Subsystem Risk Patterns

Risk Pattern	Description	Subsystems Most Affected
Critical alloy inputs	Nickel, chromium, titanium, and vanadium support multiple turbine and heat-exchange systems	Gas turbines, steam turbines, HRSG, cooling systems
Heavy manufacturing & precision manufacturing processes	Large components require specialized forging, casting, coating, and machining capacity	Turbines and other large structural components
Electrical & control equipment	Dependence on semiconductor components and specialized electrical equipment	DCS and plant electrical systems

To examine these risks in greater detail, five focus areas were selected for detailed supply-chain assessment based on criteria described in Section 3.1. These include upstream alloying inputs such as nickel and chromium; precision manufacturing of turbine blades and vanes; production of large forgings for rotor shafts, disks, and casings; and custom fabrication of finned-tube heat exchangers. These areas represent supply-chain stages where supply-chain disruptions could spread across multiple subsystems and influence power plant construction timelines.

By contrast, many structural components and standard mechanical systems rely on mature industrial supply chains with established domestic or allied manufacturing capacity and are less likely to constrain near-term deployment.

Priority Mitigation Areas

The analysis identifies mitigation pathways aligned with the critical supply-chain areas examined in this report (See Table I-3). Across the four focus areas, mitigation priorities emphasize three recurring strategies: increasing throughput in existing qualified facilities, strengthening allied supply partnerships, and stabilizing long-term demand signals to support U.S industrial investment.

Table I-3: Priority Mitigation Actions by Deployment-Critical Area

Focus Area	Key Supply-Chain Challenges	Priority Mitigation Actions
Upstream alloys (Nickel & chromium)	High foreign dependence on critical alloy inputs and limited domestic refining capacity	Expand secondary supply through scrap recycling and byproduct recovery; increase throughput at existing qualified facilities; strengthen allied sourcing partnerships
Precision manufacturing (Turbine blades & vanes)	Limited qualified casting and coating capacity and long production cycles	Increase throughput at existing qualified blade and vane facilities; diversify capital-equipment sourcing through allied co-production; support investment through federal financing tools
Heavy forging infrastructure (Disk, shaft, & casing)	Limited global capacity for very large forgings and long lead times for specialized equipment	Stabilize demand signals to support supplier investment; maximize throughput at existing domestic forging assets; leverage allied forging capacity
Custom heat-exchange fabrication (Finned tubes)	Limited finned-tube fabrication throughput and demand volatility	Stabilize HRSG demand signals to support domestic investment; secure access to allied fabrication capacity; expand domestic fabrication through targeted incentives

Scope and Future Analysis

The subsystem screening intentionally emphasized subsystems relevant across multiple power-generation technologies, including coal, geothermal, hydropower, and natural gas generation. Nuclear-specific components were excluded because they are examined in a separate study. While thirteen components were identified through system-level screening, detailed analysis was limited to five focus areas selected to align with the sprint scope and timeline.

Additional materials and components, including semiconductor wafers and chips and specialty electrical components such as transformers, also showed elevated risk signals but were not examined in detail because they are addressed in other studies.

Future work could extend the analysis to additional components identified during the screening, including technology-specific equipment for geothermal and hydropower and additional electrical, control, and advanced-material systems.

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1 Power Generation Landscape and Supply Chain Scope

The U.S. electric power sector has entered a period of sustained growth after decades of relatively steady increases in electricity demand (EIA, 2025; EIA, 2026). The rapid expansion of data centers supporting artificial intelligence (AI), cloud computing, and other digital infrastructure, combined with the reshoring and expansion of domestic manufacturing, is expected to drive significant increases in electricity consumption over the coming decade. Federal and private investments are accelerating the development of energy-intensive industries such as semiconductor fabrication, battery manufacturing, and critical minerals processing, while broader electrification trends across transportation, buildings, and industrial sectors continue to increase electricity demand. Meeting this growing demand will require substantial additions of generation capacity and supporting infrastructure while maintaining reliability, affordability, and resilience.

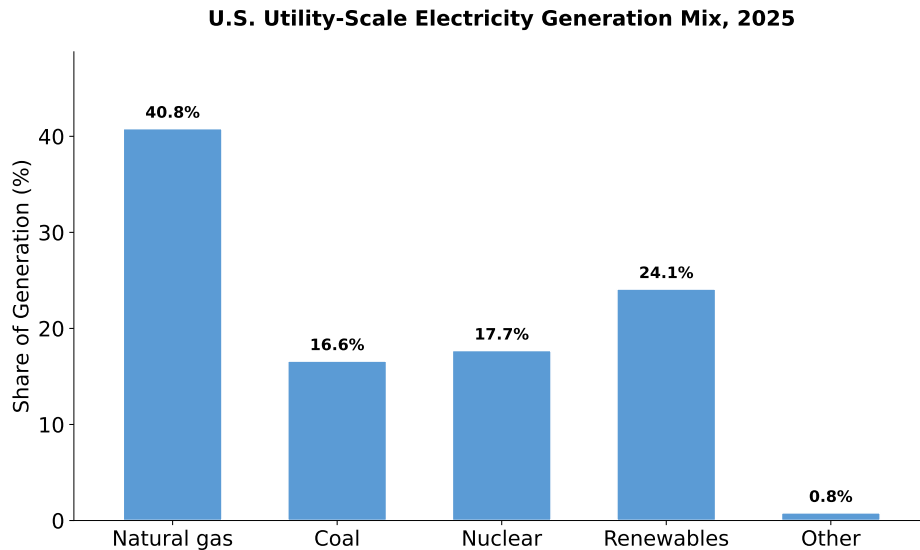


Figure 1: U.S. Utility-Scale Electricity Generation Mix

Source: U.S. Energy Information Administration (EIA), Electric Power Monthly (August 2026)

U.S. electricity supply relies on a mix of generation technologies (See Figure 1). In 2025, natural gas accounted for 40.8% of utility-scale generation, followed by renewables (24.1%), nuclear (17.7%), and coal (16.6%). While renewable energy deployment is expected to continue to grow rapidly, natural gas is projected to remain a significant part of the generation mix through at least 2030 (See Figure 2). As utilities and independent power producers expand, modernize, and maintain generation assets to meet rising demand, securing the supply chains that support these technologies is becoming increasingly important. The ability of these supply chains to provide critical equipment, materials, and services in a timely and cost-effective manner will remain important to electric system reliability, affordability, and resilience.

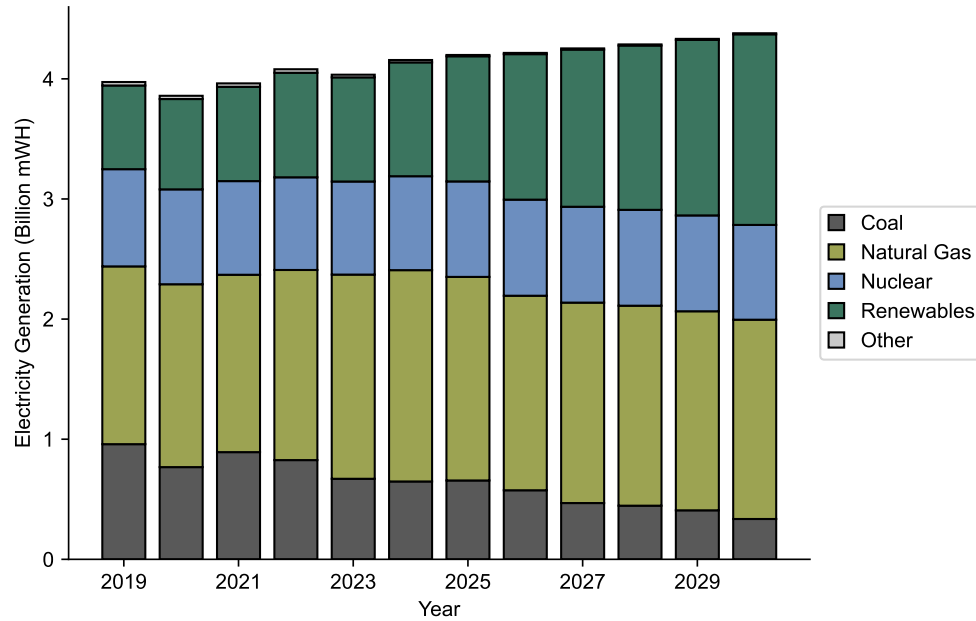


Figure 2: U.S. Electricity Generation Trends

Data from EIA Electric Power Monthly, Sept 2025; EIA Annual Energy Outlook, 2025

Note: Generation is from utilities and independent power producers

Given its current and projected role in U.S. electricity generation, NGCC was used as the reference for the detailed subsystem assessment. The broader subsystem screening, however, was structured to include systems relevant across multiple power-generation technologies. Many of the subsystems and equipment categories evaluated for NGCC are also used in coal, hydropower, geothermal, and other generation technologies. This approach allows the analysis to identify common supply-chain risks across these technologies. Nuclear remains an important source of electricity, but reactor-specific technologies and supply chains are addressed in a separate study.

2 Power-Plant Subsystems

This section defines the major subsystems that make up power-generation facilities and establishes the basis for the supply-chain assessment. Power-generation facilities are composed of multiple interconnected subsystems that perform several functions such as fuel handling, energy conversion, heat management, power generation, and system control. While configurations vary by technology, many of these subsystems are shared across natural gas, coal, hydropower, and geothermal systems. Box 2.1 presents a representative natural gas combined cycle (NGCC) configuration and identifies the eleven subsystems assessed in this report. It also shows how these subsystems apply across different generation technologies.

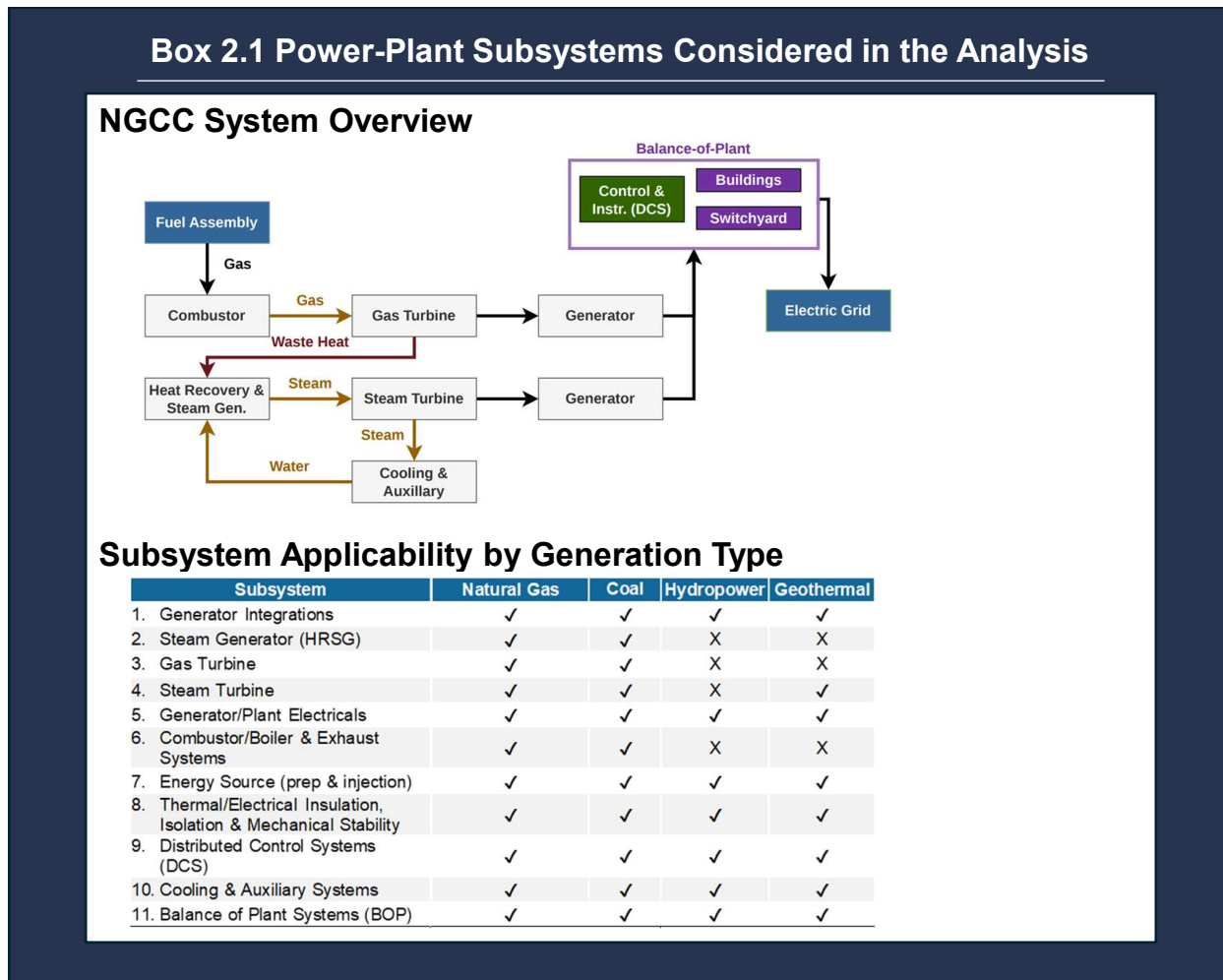


Figure 3: Box 2.1 Power-Plant Subsystems Considered in the Analysis

Across generation types, these subsystems can be grouped into four main categories:

- **Energy conversion systems:** Gas turbines, steam turbines, and associated combustion or boiler systems

- **Heat management systems:** Heat recovery & steam generation, cooling, and auxiliary systems
- **Power generation systems:** Generators, electrical systems, and facility interface connectors
- **Control and support systems:** Distributed control systems (DCS), balance-of-plant (BOP) systems, and supporting infrastructure

Hydropower systems were included due to overlaps in key supply chains such as large castings and electrical equipment. Hydro turbine manufacturing itself was not examined in detail due to long equipment lifetimes, limited new builds, and site-specific designs. This subsystem structure provides the basis for the screening framework described in the next section.

3 Supply-Chain Screening Framework

3.1 Technology and Supply Chain Screening

The assessment began with a broad review of power-plant subsystems to ensure full system coverage before narrowing the analysis to the components most likely to constrain deployment. A consistent screening framework (See Appendix A) was applied to evaluate supply-chain risks across subsystems and identify areas for deeper analysis (See Box 3.1).

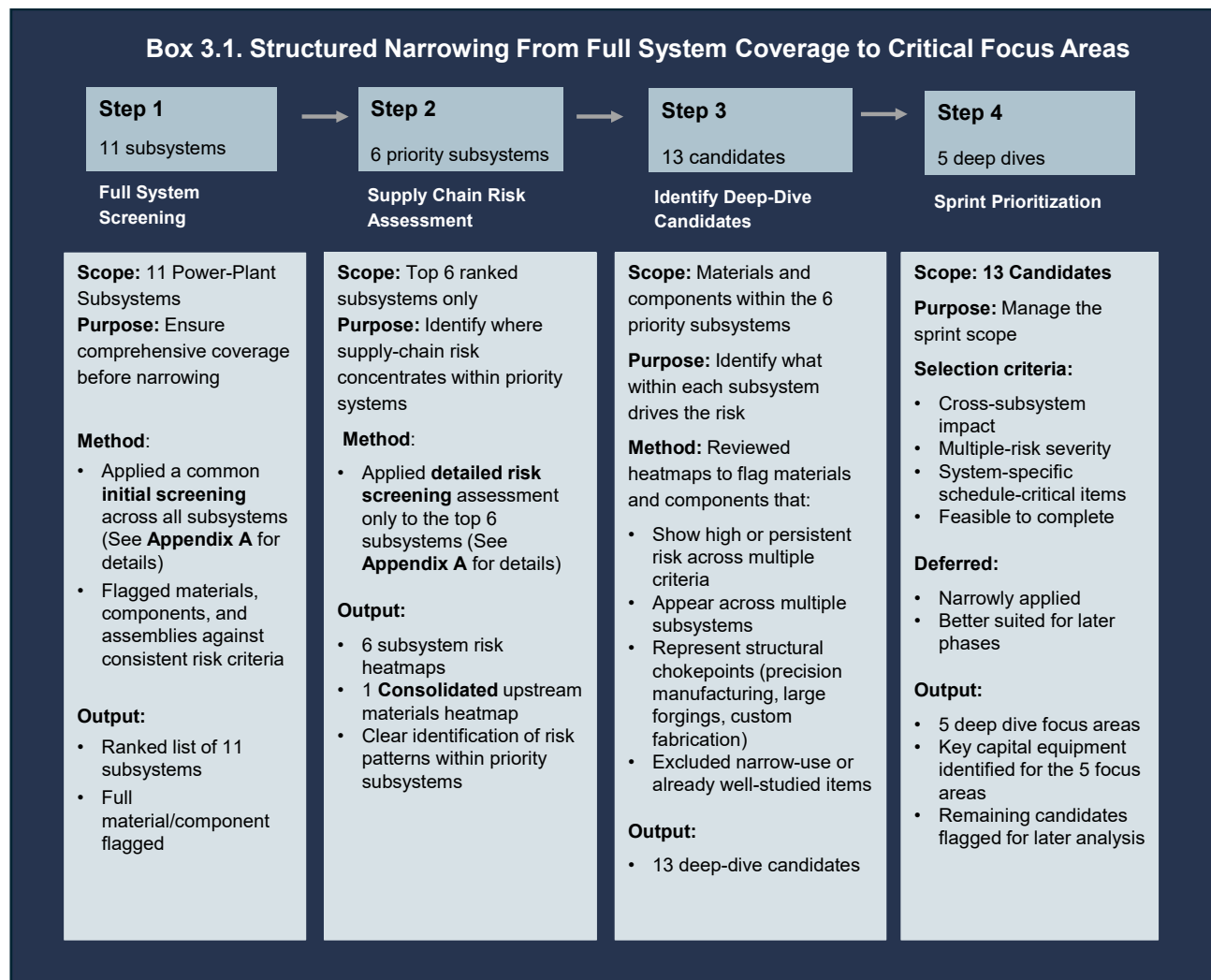


Figure 4: Box 3.1. Structured Narrowing From Full System Coverage to Critical Focus Areas

The screening framework evaluates subsystems across five criteria that capture key structural sources of supply-chain risk:

- **Cross-technology use:** Components used across multiple power plant types or other sectors

to assess competition for supply

- **Foreign input dependence:** Reliance on imported materials or components
- **U.S. manufacturing constraints:** Availability of qualified production capability
- **Critical minerals exposure:** Dependence on materials considered critical by U.S. Department of Energy (DOE) assessment (Bauer, Diana, et al. 2023)
- **Manufacturing complexity:** Specialized processes, equipment or expertise that may increase production lead times

Subsystems that scored highly across multiple criteria were prioritized for deeper analysis. The framework also considered capital equipment requirements associated with key manufacturing processes. To manage scope, detailed capital equipment assessment focused on the five priority focus areas identified through the screening process. This approach allowed the assessment to move from broad system coverage to a focused evaluation of deployment-critical supply chains.

3.2 Mitigation Pathway Prioritization

To evaluate potential responses to supply-chain risks identified in the assessment, the analysis applied six mitigation pathway categories. These categories reflect the primary mechanisms through which supply constraints may be addressed. The mitigation framework includes:

- **Domestic capacity expansion:** Increasing U.S. production capability for critical materials or components.
- **Capital equipment onshoring and diversification:** Expanding domestic manufacturing capability for specialized industrial equipment and diversifying supply sources for critical production systems.
- **Allied sourcing:** Diversifying supply through trusted international partners.
- **Design and qualification flexibility:** Reducing constraints on narrowly specified materials, components, or processes through design changes, standards updates, or qualification reforms.
- **Demand coordination and sequencing:** Aligning demand signals across sectors to support domestic investment.
- **R&D for alternative materials and processes:** Advancing alternative materials, manufacturing processes, or technologies.

Each focus area is evaluated against these mitigation pathways to identify mitigation options and highlight where different levers may reduce supply-chain risk. Assessment of the costs associated with implementing these mitigation options is outside the scope of this study.

4 Full-System Screening Outcomes

The full-system screening identified supply-chain risk signals across all eleven power-plant subsystems, confirming that exposure is system-wide and requires prioritization. Six subsystems were selected as focus areas for this sprint (See Table 4-1), based on the screening methodology described in Section 3.

Table 4-1: Subsystem Screening Outcomes

Subsystem	Screening Outcome
Gas Turbine	Deep-dive focus
Steam Turbine	Deep-dive focus
Steam Generator (HRSG)	Deep-dive focus
Distributed Control Systems (DCS)	Deep-dive focus
Cooling & Auxiliary Systems	Deep-dive focus
Balance-of-Plant Systems (BOP)	Deep-dive focus
Combustor & Exhaust Systems	Future analysis candidate
Generator Integrations	Future analysis candidate
Thermal/electrical Insulation, Isolation, & Mechanical Stability	Future analysis candidate
Fuel Assembly (preparation & injection)	Future analysis candidate
Generator/Plant Electricals	Future analysis candidate

Five additional subsystems also demonstrated meaningful supply-chain exposure but were not examined in detail during this sprint due to time and resource constraints. These areas remain important candidates for future analysis.

5 Cross-Subsystem Risk Patterns

The detailed supply risk assessment of the six prioritized subsystems reveals several recurring supply-chain patterns that appear across multiple parts of the power-plant architecture. Table 5-1 summarizes the main areas where risks concentrate as well as areas where supply chains appear comparatively resilient.

Table 5-1: Cross-Subsystem Supply-Chain Risk and Strength Patterns

Category	Pattern	Primary Risk Drivers	Affected Subsystem
High Risk Areas	Upstream Alloy Inputs	High foreign dependence on key metals (e.g. nickel, chromium, titanium, vanadium), combined with limited domestic refining capacity, and strong cross-sector demand from aerospace, batteries, defense, and advanced manufacturing	Gas turbines, steam turbines, HRSG, cooling systems, and BOP
High Risk Areas	Heavy Manufacturing and Precision Manufacturing Processes	Limited global or domestic capacity for large forgings and precision-manufactured components, combined with long production cycles, specialized capital equipment, and certification requirements	Steam turbines, gas turbines, HRSG, and other large structural plant components
High Risk Areas	Electrical and Control Equipment	Dependence on semiconductor components & equipment manufacturing, limited domestic production capacity, and long lead times for grid-interface systems	DCS, BOP electrical systems e.g., transformers, switchgears
Areas of Relative Strengths	Basic industrial metals and structural inputs	Diversified global supply and established domestic and allied production capacity for widely used industrial metals	Multiple subsystems including turbines, cooling systems, and BOP equipment
Areas of Relative Strengths	Final system fabrication and assembly	Mature OEM integration networks and established industrial fabrication capacity for plant-level assembly	Gas turbines, steam turbines, steam generators, and BOP systems
Areas of Relative Strengths	Standard industrial mechanical components	Established supplier bases and widely used manufacturing processes for pumps, casings, and auxiliary mechanical components	Cooling & auxiliary systems, BOP systems

Overall, deployment risk concentrates on a few structural areas, particularly upstream alloy inputs, heavy manufacturing and precision manufacturing processes, and specialized electrical and control equipment. Other parts of the supply chain rely on mature industrial ecosystems with broader supplier bases and established manufacturing capacity. The next section examines the thirteen components identified through the subsystem deep dives to assess the specific constraints most relevant to near-term deployment.

6 Critical Materials and Components

The subsystem assessments identified specific materials and components where supply-chain constraints could affect power-plant deployment timelines. Applying the decision logic described in Section 3.1 across the six prioritized subsystems, thirteen candidates were identified for deep dives (See Table 6-1).

Table 6-1: Deployment-Critical Components Identified from Subsystem Analysis

Category	Deep Dive Candidate	Affected Subsystems
Critical Materials and Alloys Inputs	Nickel	Gas turbine, steam turbine, HRSG, BOP, cooling & auxiliary
Critical Materials and Alloys Inputs	Chromium	Steam turbine, HRSG, BOP, cooling & auxiliary
Critical Materials and Alloys Inputs	Titanium	Steam turbine, gas turbine, cooling & auxiliary
Critical Materials and Alloys Inputs	Vanadium	Gas turbine, HRSG, BOP
Manufacturing and Components	Turbine blades & vanes	Gas turbine, steam turbine
Manufacturing and Components	Large forgings (rotor shafts, rotor disks, casings)	Gas turbine, steam turbine
Manufacturing and Components	Finned tubes	HRSG
Manufacturing and Components	High-pressure flanges	HRSG
Manufacturing and Components	Control boards	DCS
Manufacturing and Components	High-voltage relays	DCS
Manufacturing and Components	Fluoropolymers	BOP (GIS, breakers, insulation)
Manufacturing and Components	Copper-nickel alloys	Cooling & auxiliary
Manufacturing and Components	Heat exchangers	Cooling & auxiliary

Several additional materials and components also showed high screening scores but were not included in the thirteen components prioritized for detailed assessment. Materials such as polysilicon, tantalum, niobium, and fluorine exhibit elevated supply-chain risk but are narrowly applied within specific subsystems. Semiconductor wafers and chips, Grain Oriented Electrical Steel (GOES), circuit breakers, transformers also ranked highly but were excluded because they are either widely

studied or covered by other studies.

To manage the scope of the sprint, five components were selected for detailed analysis using the criteria described in Section 3.1. The remainder of the report examines these components across the following four main sections:

- **Upstream Materials:** Nickel and chromium
- **Precision Manufacturing:** Turbine blades and vanes
- **Heavy Forging:** Large forgings relevant for rotor shafts, rotor disks, and casings
- **Custom Heat-Exchange Fabrication:** Finned tubes

7 Upstream Materials: Nickel and Chromium

Nickel and chromium serve as foundational inputs for multiple subsystems. These materials enable stainless steels, specialty steels, and superalloys used in turbines, steam generators, cooling systems, and balance-of-plant equipment (Nickel Institute, n.d; Bauer et al., 2023).

7.1 Nickel and Chromium: Manufacturing Bottlenecks

For nickel, key bottlenecks occur in smelting and refining. Expanding U.S. refining capacity faces several challenges, including cost competitiveness and the technical complexity of nickel processing. Some refining routes require harsh operating conditions and specialized equipment, contributing to higher processing costs (Li et al. 2025). Box 7.1 below shows how both materials flow from ore extraction and refining into alloy production and ultimately into multiple plant subsystems.

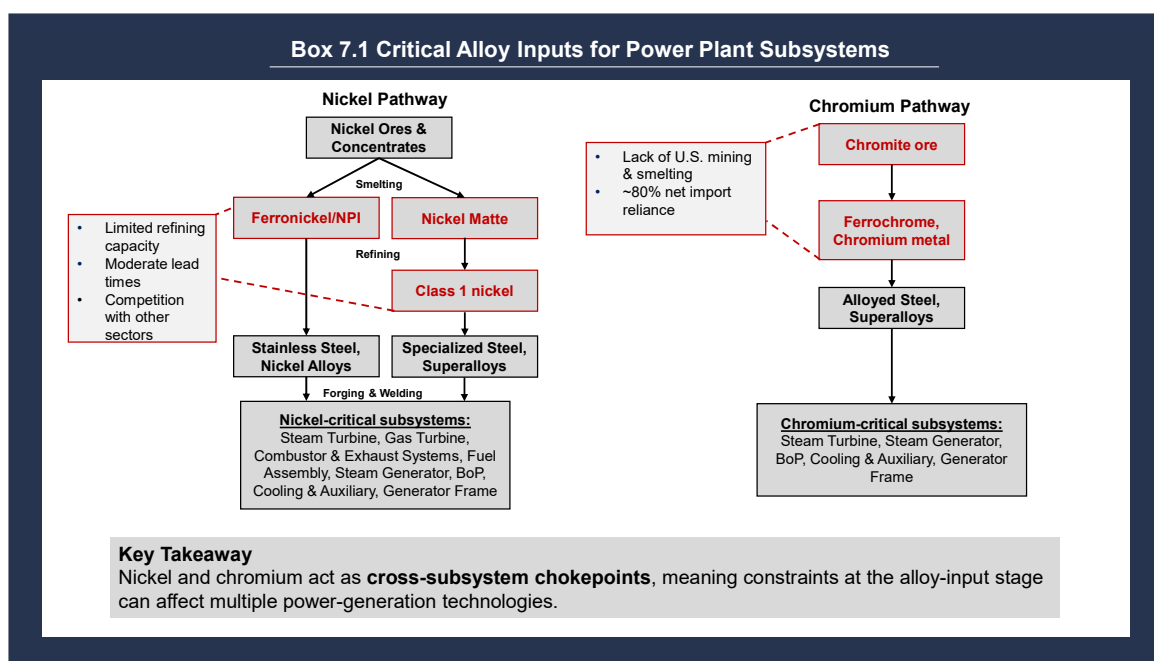


Figure 5: Box 7.1. Nickel and Chromium Supply and Manufacturing Flow

For chromium, bottlenecks are on chromite ore and ferrochromium. The U.S. has limited reserves, and has no primary mining and smelting capacity, while global supply is geographically concentrated: Chromite production in South Africa and ferrochromium refining in the People's Republic of China (PRC), contributing to supply exposure and moderate lead times (USGS 2025).

7.2 Nickel and Chromium: Global Supply Concentration and Deployment Risk

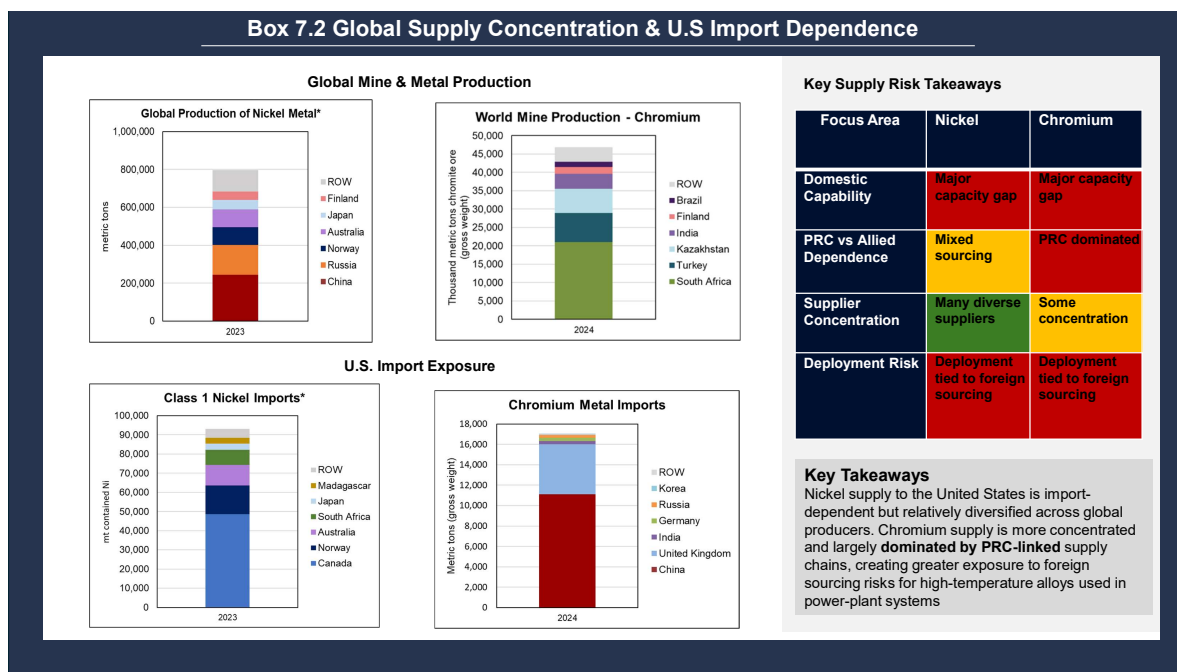


Figure 6: Box 7.2. Nickel and Chromium Global Supply Concentration

Nickel metal production and import data is from the most recent (2023) USGS mineral yearbook. Nickel metal refers to Class 1 nickel, which has a purity greater than 99%.

Nickel and chromium supply chains exhibit significant global concentration (See Box 7.2), increasing exposure to geopolitical disruptions, price volatility, and long lead times. Chromium presents the highest risk: global mine production and refining are concentrated in a small number of countries, and the U.S. remains highly import-reliant (USGS, 2024). Nickel sourcing is relatively more diversified, with the most U.S. imports supplied by allied countries.

7.3 Nickel and Chromium: Mitigation Pathways

Mitigation pathways were screened using the mitigation pathways from Section 3.2 to identify actions that can reduce deployment risk associated with nickel and chromium supply constraints. For these materials, mitigation strategies that expand domestic material availability or diversify sourcing are most relevant. Other levers, such as design flexibility or demand coordination, have more limited applicability for globally traded alloy inputs (See Box 7.3.1).

Box 7.3.1 Nickel & Chromium: Mitigation Lever Relevance

Mitigation Levers	Relevance	Rationale
Domestic capacity expansion	✓	Expansion of domestic mining, refining, and recycling reduces import reliance
Capital equipment onshoring & diversification	✗	Outside scope; focus area addresses material supply rather than manufacturing equipment
Allied sourcing	✓	Diversifying sourcing across allied suppliers reduces concentration and geopolitical risk
Design & qualification flexibility	✓	Limited near-term applicability; longer-term relevance through alloy optimization and broader material qualification
Demand coordination & sequencing	✗	Limited applicability for globally traded nickel markets. Demand coordination may support domestic production of specialty alloy feedstocks or high-purity nickel used in superalloys
R&D for alternative materials & processes	✓	Materials innovation and process improvements may reduce dependence on constrained alloy inputs

Figure 7: Box 7.3.1. Nickel and Chromium Mitigation Pathway Screening

Current mitigation efforts focus primarily on expanding domestic supply capacity and advancing alternative production pathways through recycling and process innovation. However, structural constraints remain, particularly the absence of domestic primary Class 1 nickel refining capacity and continued reliance on imported chromium intermediates (See Box 7.3.2). High-purity requirements for turbine and generator applications also limit the use of conventional steel scrap. Advanced recycling approaches that enable ultra-high purity or prioritize superalloy-grade recovery may help address these constraints.

Box 7.3.2 Nickel & Chromium: Current Mitigation Landscape

Relevant Mitigation Levers	Existing Actions	Remaining Gaps
Domestic capacity expansion	Talon Metals: Defense Production Act (DPA) Title III (\$20M) for nickel exploration + BIL(\$115M) for mineral processing facility; and acquired Eagle Mine with plan for expansion (<i>Talon Metals Corp. 2025a, 2025b</i>) 6K Additive: DPA Title III (\$23.4M) expand metal production from recycled feedstocks (e.g., Nickel Alloys 635/718, SS316, & high purity chromium powder) (<i>6K Additive n.d.</i>) Fixing America's Surface Transportation Act Title 41 (FAST-41) added designation for Nikolai Mine(AK), has significant chromium resources (<i>Permitting Council n.d</i>)	- No creation of domestic primary Class 1 nickel refining capacity - Does not eliminate reliance on imported ores and intermediates
Allied sourcing	Class 1 nickel imports largely sourced from allies , especially Canada (<i>USGS 2025</i>) Chromium imports are highly concentrated: roughly 50% of U.S. ferrochromium imports come from South Africa, while more than half of chromium metal imports originate from China(<i>USGS 2025</i>)	- Chromium metal dependence to PRC remains high with limited diversification Continued exposure to global price volatility and supplier queuing
R&D for alternative materials & processes	DOE (AMMTO, ARPA-E), DOW and national labs supporting alternative production pathways like tailings extraction and advanced recycling, e.g., Talon granted \$2.5M from Defense Logistic Agency (DLA) for pilot-scale testing (<i>Talon Metals Corp. 2025a</i>)	- Limited near-term deployment impact - Long qualification and scale-up timelines

Figure 8: Box 7.3.2. Nickel and Chromium Current Mitigation Landscape

Building on the current mitigation landscape, near term options could include expanding secondary and unconventional supply, increasing throughput at existing facilities, and strengthening allied sourcing to reduce exposure to concentrated supply chains. Medium-term options could include development of domestic processing and refining capacity, while longer-term strategies focus on R&D and alternative production pathways (See Box 7.3.3).

Box 7.3.3 Nickel & Chromium: Prioritized Mitigation Pathways	
Relative Impact	Mitigation Options & Illustrative Examples
Near-term impact	Expand secondary and unconventional nickel and chromium supply: Scale superalloy and stainless scrap recycling, byproduct recovery, and tailings-based extraction where technically viable, including through public-private partnerships. Examples of levers include national lab-led consortia, DPA Title III awards, and DOE grant support. Increase throughput at existing qualified facilities: Extend mine life/footprint (e.g., Talon Metals at Eagle Mine) and expand recycling-based alloy production through initiatives like 6K additive Strengthen allied sourcing: Deepen partnerships with allied suppliers to diversify supply routes and reduce exposure to single-country disruptions, including use of financing and trade facilitation tools (e.g., EXIM and DFC – type support) where appropriate
Mid-term impact	De-risk domestic processing and refining capacity: Support processing and refining projects using existing authorities (e.g., DPA Title III, DLA, DOE programs) to reduce post-import bottlenecks Advance mining projects delayed by permitting: Utilization of FAST-41 program and similar mechanisms to accelerate mineral projects stalled by multi-year permitting processes
Longer-term impact	Advance R&D and alternative production pathways: Support advanced recycling, alternative extraction methods, and material substitutions through DOE, Department of War (DoW), and national lab programs where qualification pathways exist Advanced exploration & mines development where economically viable, recognizing long lead times, permitting hurdles, capital intensity, and limited near-term deployment impact for nickel and chromium
Note: Relative impact reflects a qualitative assessment of when each mitigation option is likely to begin meaningfully influencing the identified supply-chain constraint, considering practical implementation timelines	

Figure 9: Box 7.3.3. Nickel and Chromium Priority Mitigation Actions

8 Precision Manufacturing: Blades and Vanes

Turbine blades and vanes are precision hot-section components that convert fluid energy into mechanical rotation and guide flow across turbine stages. Their complex geometries, internal cooling channels, and tight manufacturing tolerances make them among the most manufacturing-intensive components in turbine systems (Juliano et al. 2024; Angel and Basak 2020).

8.1 Blade and Vanes: Manufacturing Bottlenecks

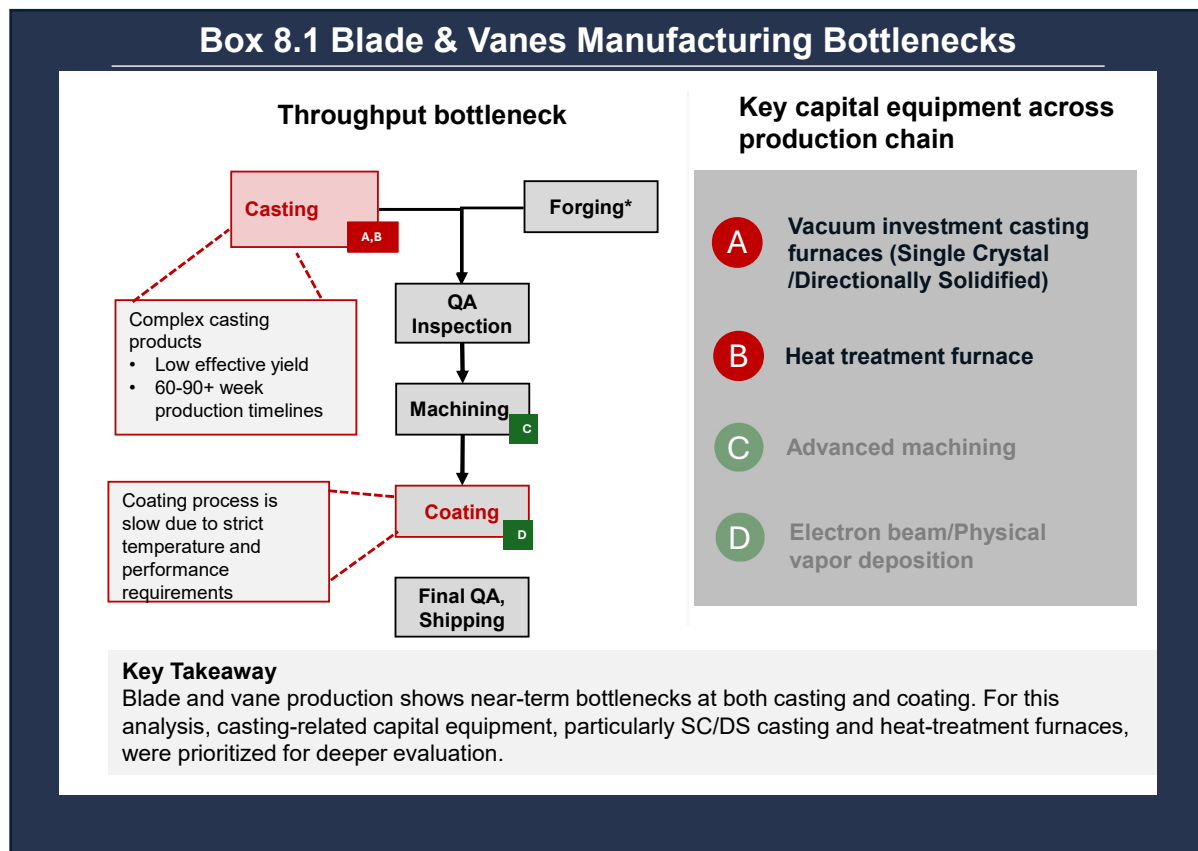


Figure 10: Box 8.1. Turbine Blade and Vane Manufacturing Bottlenecks

Production bottlenecks occur primarily in precision investment casting (See Box 8.1), where complex geometries reduce effective yields and extend manufacturing timelines to 60–90+ weeks for most advanced castings (Angel and Basak 2020), placing blade and vane production among the longest-lead components in turbine assembly. Coating and finishing steps introduce additional

constraints due to strict temperature control requirements, specialized equipment, the need for rare earth elements (e.g., Yttrium) and OEM qualification standards (Juliano et al. 2024). These production constraints are reinforced by limited availability of specialized capital equipment like single-crystal/directionally solidified (SC/DS) casting furnaces and high-uniformity vacuum heat-treatment furnaces. SC/DS systems require high capital investment, long installation timelines, OEM-specific process qualifications, and proprietary know-how concentrated among a small number of foreign equipment manufacturers (ALD Vacuum Technologies n.d., ASM International n.d., Growth Market Reports n.d.).

8.2 Blade and Vanes: Global Supply Concentration and Deployment Risk

Direct public data on turbine blade and vane production is limited. As a result, this analysis relies on proxy indicators including investment casting market share, representative U.S. blade manufacturing capacity, and turbine component trade flows (See Box 8.2). Evidence suggests the U.S. maintains significant domestic blade and vane production capability with major OEM facilities operated by GE, Siemens, and Mitsubishi Power performing much of the blade manufacturing cycle in-house from casting through final inspection and finishing (Gibson 2024, Turboblading Group n.d.). Also, North America holds a leading share of high-value precision casting capacity used across aerospace, defense, and industrial turbine applications (Sullivan and Muxlow 2025). Trade data indicate that turbine parts and components are sourced from allied countries such as Mexico, Japan, and several European suppliers. Alloy inputs are also largely sourced from allied producers in Europe and Australia (U.S. International Trade Commission Dataweb, n.d.). However, casting and heat treatment equipment used in blade manufacturing remains concentrated among specialized foreign manufacturers, primarily in Europe and Japan. These furnace systems are highly specialized systems integrated into OEM production processes, limiting the ability to rapidly expand production capacity. Precision casting capacity and specialized furnace availability present a greater deployment risk than foreign component sourcing.

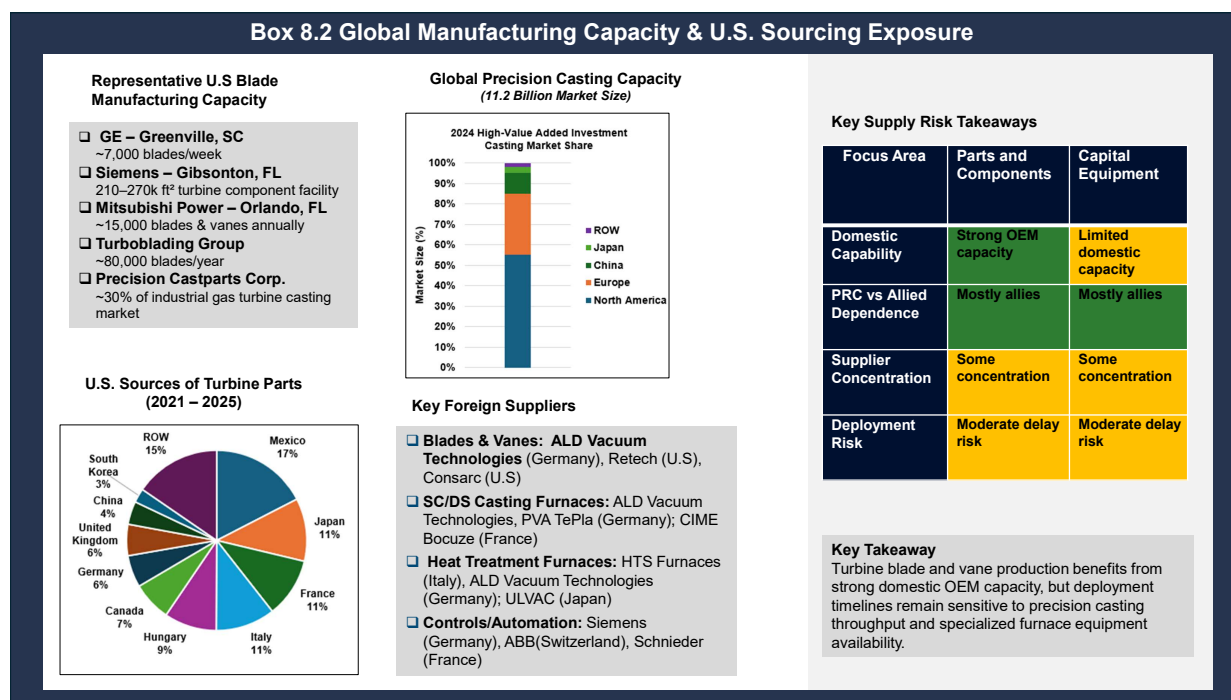


Figure 11: Box 8.2. Turbine Blade and Vane Global Supply and Capacity Indicators
 Chart of imports aggregates steam (HS 84069090) and gas turbine parts (HS 84119900) (U.S. International Trade Commission, n.d.; Sullivan, Tim, and Merrin Muxlow. 2025).

8.3 Blade and Vanes: Mitigation Pathways

Given the structure of blade and vane supply chains, mitigation strategies focus on three core challenges: limited precision casting throughput, constrained availability of specialized furnace equipment, and long capacity-expansion timelines. Thus, the most relevant mitigation levers could focus on expanding casting capacity, diversifying access to critical capital equipment, and strengthening demand coordination across turbine markets (see Box 8.3.1).

Box 8.3.1 Blades & Vanes: Mitigation Lever Relevance		
Mitigation Levers	Relevance	Rationale
Domestic capacity expansion	✓	Expanding blade casting capacity can reduce component lead times
Capital equipment onshoring & diversification	✓	SC/DS casting furnaces and controls are largely supplied by foreign manufacturers
Allied sourcing	✗	Sourcing already relies primarily on allied suppliers
Design & qualification flexibility	✗	Blade materials and processes are constrained by strict performance and qualification requirements
Demand coordination & sequencing	✓	Casting capacity also serves aerospace demand, creating competing production schedules
R&D for alternative materials & processes	✓	Improvements in casting and manufacturing processes can increase throughput and reduce costs

Figure 12: Box 8.3.1. Blade and Vane Mitigation Pathway Screening

Industry and policy efforts already address elements of these challenges. However, production capacity remains constrained by long equipment lead times and concentrated capital-equipment manufacturing (see Box 8.3.2).

Box 8.3.2 Blades & Vanes: Current Mitigation Landscape		
Relevant Mitigation Levers	Existing Actions	Remaining Gaps
Domestic capacity expansion	- SC/DS casting and heat-treatment furnaces operated by turbine OEMs and aerospace suppliers e.g., GE Vernova, Siemens, and Mitsubishi Throughput increases are underway at existing sites through process improvements and equipment expansion(<i>DOW 2024; Reuters. 2025; SECO/WARWICK n.d.; Utility Dive 2025</i>)	- Capacity expansion constrained by long lead times (>1.5 years) for new furnaces - Limited build slots and specialized commissioning slow near-term scale-up
Capital equipment onshoring / diversification	- Furnace engineering, integration, and servicing capability exists in the U.S. e.g., Retech, Consarc (<i>ASM International n.d.; Inductotherm Group n.d.</i>) - Some U.S. manufacturing of furnace subsystems exists, but are concentrated amongst few suppliers (<i>Retech discovery call</i>)	- Capital equipment still concentrated among a small number of suppliers - Core furnace technologies, controls, and vacuum systems remain foreign-dominated - Key subcomponents still imported, maintaining exposure to supplier queues
Demand coordination / sequencing	No known coordinated, cross-sector, or policy-driven demand smoothing or sequencing mechanisms for blades and vanes (or turbine production more broadly)	- Competing demand from aerospace, defense, and industrial turbine continues to crowd limited furnace availability - Lack of sustained, coordinated demand may limit supplier investment
R&D for alternative materials & processes	DOE (AMMTO, ARPA-E), DOW, and national labs support development of advanced materials and improvement of mold making and investment casting processes (<i>ARPA-E n.d.; DOW 2024</i>)	Limited near-term deployment impact, short-term capacity expansion remains challenging

Figure 13: Box 8.3.2. Blade and Vane Current Mitigation Landscape

Based on the identified remaining gaps, Box 8.3.3 summarizes prioritized mitigation pathways. Key actions could include increasing throughput at existing qualified facilities, diversifying access to specialized furnace equipment, and improving long-term demand signals to support supplier investment.

Box 8.3.3 Blades & Vanes: Prioritized Mitigation Pathways	
Relative Impact	Mitigation Options & Illustrative Examples
Near-term impact	Increase throughput at existing qualified facilities: Engage key stakeholders involved in blade and vanes to identify incentives that will encourage capacity increases at OEM and aftermarket sites through equipment upgrades, process optimization, and shift expansions Diversify capital-equipment sourcing through allied co-production: Where full onshoring is not feasible, support partnerships between U.S. OEMs and allied equipment suppliers through policy tools such as EXIM and DFC financing, including co-production and domestic manufacturing of key equipment components
Mid-term impact	De-risk capital equipment investment through federal financing tools: Use existing authorities (e.g., DPA Title III, DOE loan and grant programs) to lower risk for furnace OEMs and large casting suppliers; and target projects that expand build slots or shorten delivery timelines Establish demand-signaling mechanisms across the aerospace and power-sector industries: Bring these industries together, using antitrust-compliant structures and appropriate legal safeguards, to provide clearer, sustained demand visibility that supports supplier investment decisions.
Longer-term impact	Advance R&D to improve yields and reduce cycle times: Continue to support improvements in mold manufacturing, investment casting processes, and heat-treatment efficiency, with emphasize of demonstration and scaling at domestic facilities Support next-generation equipment and processes via public-private partnerships: Programs for specialized industrial equipment initiatives (Heavy Press Program) can advance technology while maintaining domestic capability
Note: Relative impact reflects a qualitative assessment of when each mitigation option is likely to begin meaningfully influencing the identified supply-chain constraint, considering practical implementation timelines	

Figure 14: Box 8.3.3. Blade and Vane Priority Mitigation Actions

9 Heavy Forging: Large Forging

Large forgings are critical structural components used in turbine rotor disk, rotor shaft and casings. Their size, weight, and metallurgical requirements make them among the most capital-intensive components in power-generation supply chains.

9.1 Large Forging: Manufacturing Bottlenecks

Production bottlenecks occur primarily at the forging press stage (See Box 9.1). Very large forging presses are rare, and capital-intensive assets concentrated in a small number of facilities worldwide, thus limiting throughput and increasing lead times for large forging components (DOW 2018; S&P Global 2025; World Nuclear Association 2026). These constraints are reinforced by specialized capital equipment requirements across the production chain. The analysis prioritizes ingot-forming furnaces – Electroslag Remelting (ESR) and Vacuum Arc Remelting (VAR) and open-die forging presses. These systems require substantial capital investment and specialized engineering expertise which present significant barriers to rapid capacity expansion (Credence Research n.d.; Verified Market Research n.d.; U.S. Department of Energy 2022).

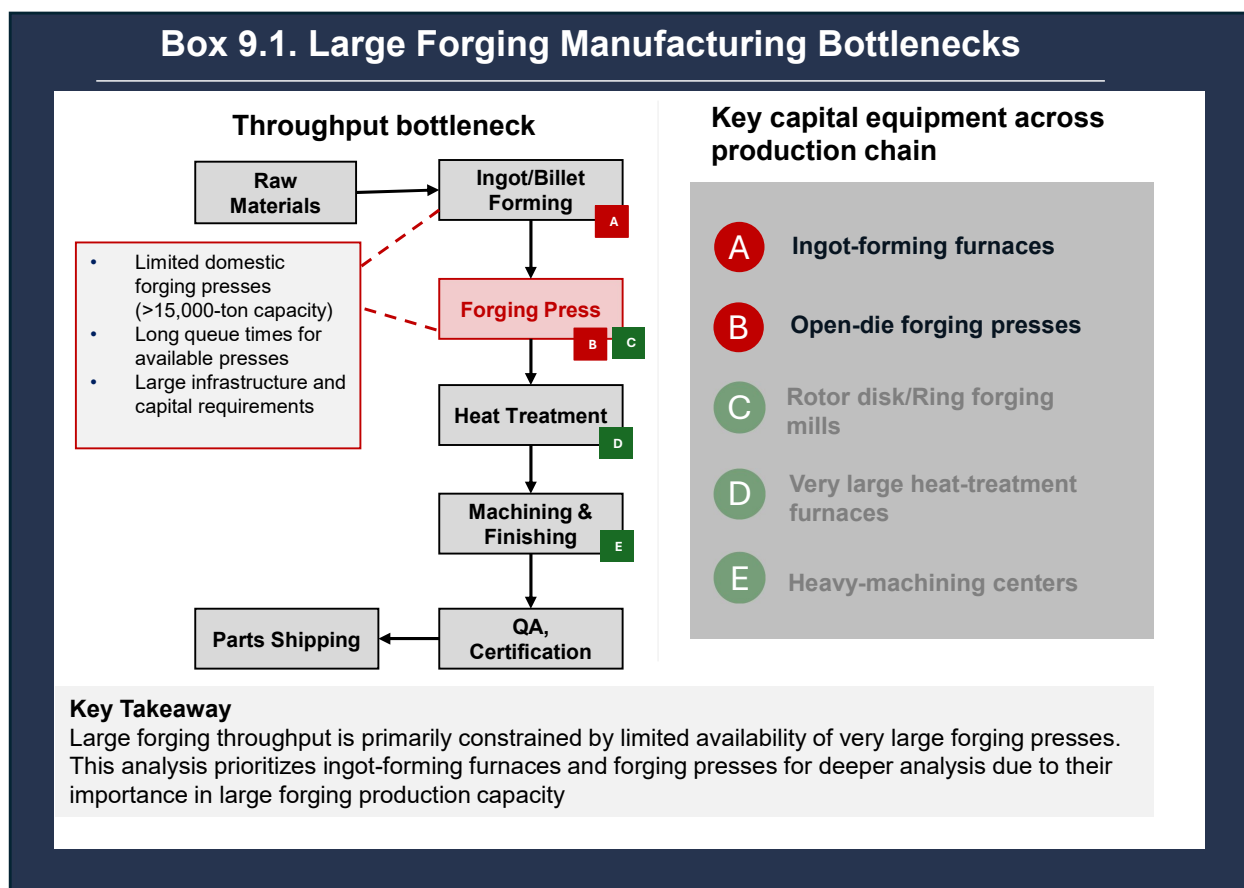


Figure 15: Box 9.1. Large Forging Manufacturing Bottlenecks

9.2 Large Forging: Global Supply Concentration and Deployment Risk

Public data on production volumes for large forged turbine components (rotor disks, rotor shafts, and casings) and associated capital equipment is limited. Global manufacturing capacity is therefore assessed using proxy indicators, including regional forging market share, representative domestic facilities, and sourcing patterns for forging equipment and suppliers.

As illustrated in Box 9.2, U.S. large-forging capability is concentrated in a small number of facilities, including Ellwood City Forge, North American Forgemasters, Lehigh Heavy Forge, and Scot Forge, which operate open-die presses capable of producing very large forgings (Ellwood City Forge Group 2025; Lehigh Heavy Forge n.d.; Scot Forge n.d.). Globally, the forging industry was estimated at roughly \$100 billion in 2026 (Industry Research 2025). By 2023, the United States accounted for approximately 14.5% of global market share (Industry Research 2025). However, heavy forgings used in large turbine components represent a small but highly specialized segment. Much of the heavy-forging capacity capable of producing very large components is concentrated in Asia-Pacific

and Europe (Alpern, 2009; Intradefairs 2025; World Nuclear Association 2025; PW Consulting 2026).

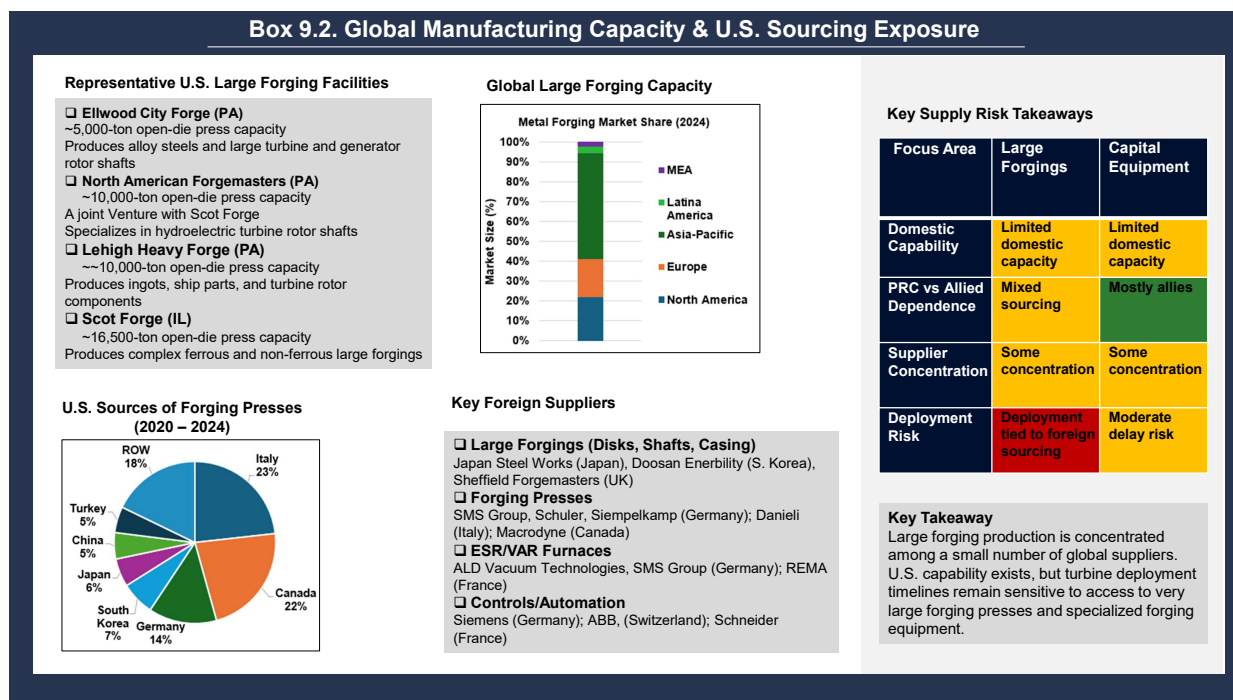


Figure 16: Box 9.2. Large Forging Global Supply and Capacity Indicators

Forging-press chart aggregates all press technologies, including open-die presses (Grand View Research 2024; UN Comtrade n.d.)

The large-forging supply chain also depends on specialized capital equipment like forging presses and ESR/VAR furnaces that are supplied by a limited number of manufacturers, including ALD Vacuum Technologies, Consarc (Inductotherm Group), SMS Group, and Schuler Group. These patterns illustrate how global capacity distribution and access to specialized forging equipment can influence the availability of large forgings.

9.3 Large Forging: Mitigation Pathways

Mitigation strategies focus on three core challenges: limited availability of very large forging presses, long lead times for new press installations, and concentrated supply of specialized melt-shop and forging equipment.

The most relevant mitigation levers could therefore focus on expanding domestic forging capacity, improving access to specialized capital equipment, leveraging allied forging suppliers, and strengthening demand coordination across sectors that rely on large forged components (See Box 9.3.1).

Box 9.3.1. Large Forging: Mitigation Lever Relevance		
Mitigation Levers	Relevance	Rationale
Domestic capacity expansion	✓	Expanding domestic forging press capacity increases throughput for large turbine forgings
Capital equipment onshoring & diversification	✓	High long-term relevance; forging press and melt-shop capacity are root constraints but require multi-year build and qualification timelines
Allied sourcing	✓	Large forging capacity is concentrated among allied suppliers, making coordinated sourcing an effective near- to medium-term option
Design & qualification flexibility	✗	Limited impact; turbine rotor materials and forging specifications are tightly constrained and cannot offset scarce large-press capacity
Demand coordination & sequencing	✓	Forging presses serve multiple sectors, coordinated demand signals may help manage deployment queues
R&D for alternative materials & processes	✓	Advanced forging processes and materials could improve throughput and reduce defects

Figure 17: Box 9.3.1. Large Forging Mitigation Pathway Screening

Industry and policy actions already address elements of these challenges. However, expansion of very large forging capacity remains constrained by long equipment lead times, high capital requirements, and fragmented demand signals across energy, defense, and industrial sectors (See Box 9.3.2).

9.3.2. Large Forging: Current Mitigation Landscape		
Relevant Mitigation Levers	Existing Actions	Remaining Gaps
Domestic capacity expansion	DOW awards supporting incremental heavy-forging throughput e.g., Lehigh Heavy Forge \$15M (<i>U.S. Department of War 2024</i>) Private investment expanding domestic capacity e.g., US Forged Rings \$700M facility (<i>Wind Systems Staff 2024</i>) Joint ventures and consortia to better utilize existing press assets e.g., North American Forgemasters; U.S. Reactor Forging Consortium (<i>Ellwood Group 2014; NuScale Power 2022</i>)	Expansion limited to throughput at existing presses; Limited domestic capacity for very large open-die presses (>10k-ton) Note: Three are conflicting industry views on available capacity; forging firms report underutilized capacity due to uncertain or fragmented demand signals and foreign competition, while turbine OEMs often report limited availability for very large forgings; improved demand coordination could help reconcile these claims (<i>Forging Industry Association 2024a; Forging Industry Association 2024b; National Defense Magazine 2024</i>)
Capital equipment onshoring & diversification	No known formal initiatives to onshore large forging presses or associated melt-shop equipment	Very large forging presses and melt-shop systems largely supplied by foreign OEMs ; long lead times and high capital costs limit rapid capacity expansion
Allied sourcing	Large presses, ESR/VAR furnaces, and automation primarily sourced from allied suppliers (Europe, Canada, Japan)	Supplier base remains highly concentrated even among allied producers; Capacity shared with aerospace and defense, sustaining queue risk
Demand coordination & sequencing	Limited, project-specific coordination between industry players to secure large forgings demand e.g., Scot Forge & Ellwood group; U.S. Reactor Forging Consortium	No formal, cross-sector or policy-driven demand coordination mechanism; Fragmented demand signals continue to deter large capital investments or expansion efforts

Figure 18: Box 9.3.2. Large Forging Current Mitigation Landscape

Building on the current mitigation landscape, key actions could include stabilizing long-term demand signals to support supplier investment, maximizing throughput at existing qualified forging facilities, and strengthening access to allied forging capacity and capital-equipment supply chains (See Box 9.3.3).

Box 9.3.3. Large Forging: Prioritized Mitigation Pathways	
Relative Impact	Mitigation Options & Illustrative Examples
Near-term impact	Stabilize investment signals to support capacity expansion: Introduce formal demand coordination mechanisms across transportation, energy, and defense sectors to provide clearer, multi-year demand signals for large forgings. This could help reconcile conflicting industry claims regarding available capacity and enable suppliers to justify incremental press upgrades or scheduling prioritization
	Maximize throughput at existing domestic forging assets: Facilitate industry–government engagement to optimize utilization of already qualified presses (e.g., Lehigh Heavy Forge, Scot Forge, Ellwood Group) to increase output without building new ultra-large presses
Mid-term impact	Leverage allied forging capacity and capital equipment supply chains through structured partnerships: Use EXIM, DFC, and DPA tools to secure long-term access to allied large-forging capacity and specialized equipment supply (e.g., presses, ESR/VAR furnaces) through offtake agreements, co-production, and coordinated contracting
	De-risk expansion of heavy-forging capacity and capital equipment through federal tools: Use DPA Title III, tax incentives, and DOE/DOD financing to support press upgrades, melt-shop capacity, and domestic production of specialized forging equipment
Longer-term impact	Advance R&D for alternative forging approaches and materials: Support research on advanced forging processes, improved alloys, and hybrid manufacturing techniques to improve throughput and reduce defects
	Explore next-generation press technologies via public–private partnerships: Long-term initiatives (e.g., Heavy Press Program–style models) to rebuild domestic capability for >10k-ton open-die presses, recognizing multi-year timelines and high capital intensity
Note: Relative impact reflects a qualitative assessment of when each mitigation option is likely to begin meaningfully influencing the identified supply-chain constraint, considering practical implementation timelines	

Figure 19: Box 9.3.3. Large Forging Priority Mitigation Actions

10 Custom Heat-Exchange Fabrication: Finned Tubes

Finned tubes are a key heat-transfer component in heat recovery steam generators (HRSGs) used in combined-cycle power plants. These tubes transfer heat from gas-turbine exhaust gases to generate steam for additional power production (Jouhara et al. 2018; GE Vernova n.d.). Finned tubes are typically assembled into tube panels and harp structures, which are then integrated into larger HRSG modules during plant construction (Buecker n.d). While finned tubes are common in industrial heat exchange, HRSG systems require a specialized fabrication segment. These units utilize high-chromium alloys such as Grade 91 alloyed steel to produce T91/P91 tubes, which are engineered to meet stringent high-temperature and pressure-part requirements under boiler codes (Swanekamp 2004).

10.1 Finned Tubes: Manufacturing Bottlenecks

Finned-tube supply chains are constrained by both material supply and fabrication capacity (See Box 10.1). Production of seamless alloy tubes used in high-temperature pressure sections requires specialized metallurgy and qualified tube mills. Fabrication requires dedicated finning lines, certified welding procedures, and heat-treatment infrastructure capable of processing large tube panels and harp assemblies.

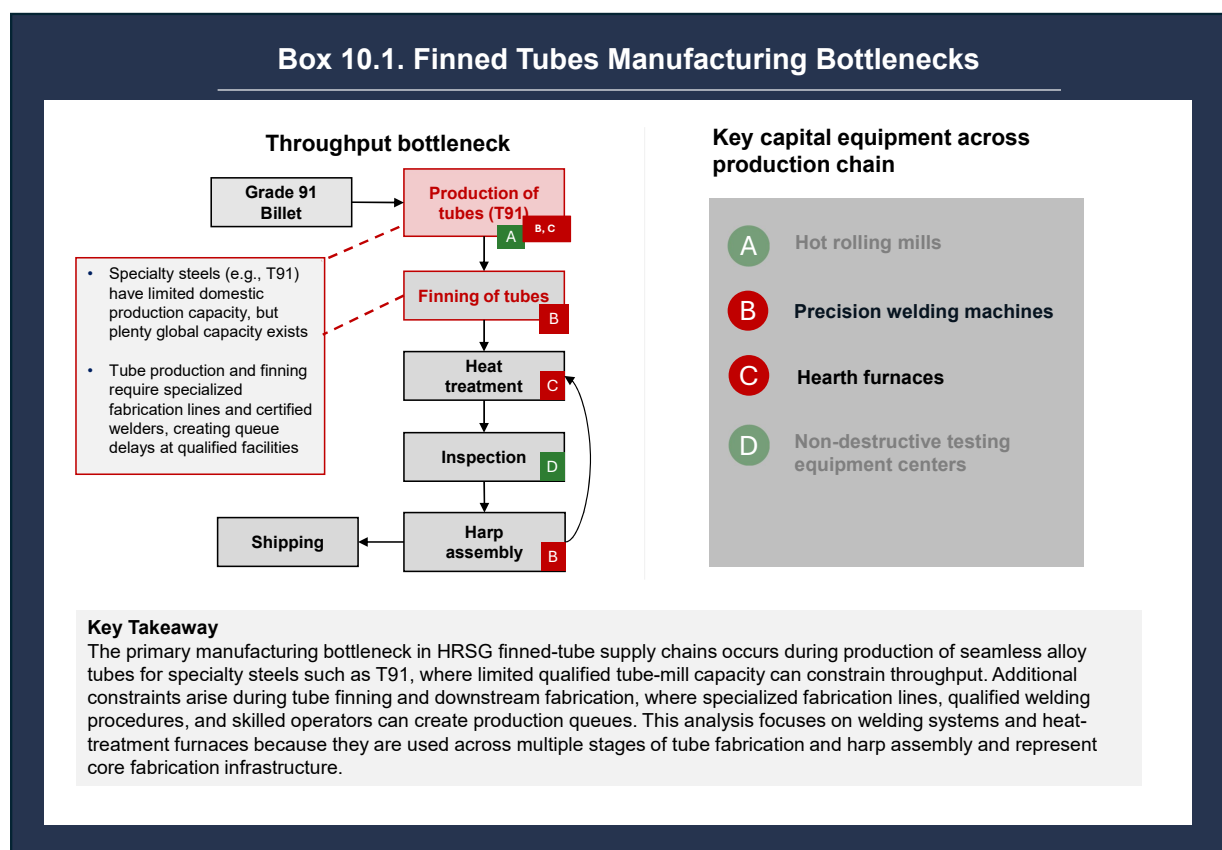


Figure 20: Box 10.1. Finned Tube Manufacturing Bottlenecks

10.2 Finned Tubes: Global Supply Concentration and Deployment Risk

Beyond fabrication constraints, the geographic structure of HRSG supply chains also shapes deployment risk. Domestic fabrication capability exists in the U.S, but large HRSG components are frequently produced through global OEM fabrication networks. These production systems are typically vertically integrated and concentrated in Asia-Pacific manufacturing hubs, particularly in South Korea, Vietnam, and Japan. As a result, U.S. buyers often rely on a mix of domestic and international suppliers for HRSG components (See Box 10.2).

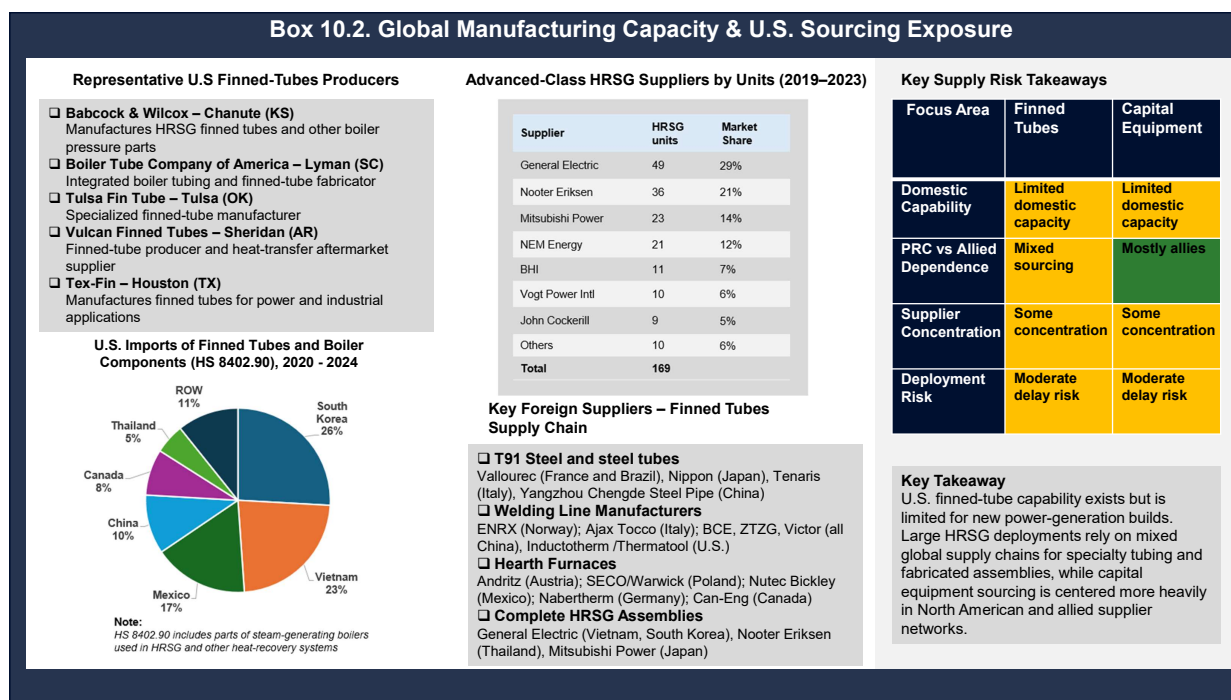


Figure 21: Box 10.2. Finned Tube Global Supply and Fabrication Network

10.3 Finned Tubes: Mitigation Pathways

Given the structure of HRSG supply chains, mitigation strategies focus on three core challenges: limited domestic fabrication scale, globally concentrated production networks, and volatility in HRSG project demand.

Potential mitigation levers could include expanding domestic fabrication capacity, coordinating demand signals across power-generation projects, and maintaining access to allied fabrication networks. Other levers, such as design flexibility or capital equipment supply, appear less central to addressing the primary constraints (See Box 10.3.1)

Box 10.3.1. Finned Tubes: Mitigation Lever Relevance		
Mitigation Levers	Relevance	Rationale
Domestic capacity expansion	✓	Expanding U.S. finned-tube fabrication capacity increases domestic throughput for HRSG heat-exchange components
Capital equipment onshoring & diversification	✗	Welding and heat-treatment equipment are available from multiple global suppliers and are not the primary constraint
Allied sourcing	✓	HRSG fabrication capacity is concentrated among vertically integrated OEM supply chains, many with Asia-Pacific production; securing and diversifying sourcing across allied suppliers could reduce allocation risk during demand surges
Design & qualification flexibility	✗	Boiler codes and OEM specifications tightly constrain HRSG materials and fabrication standards
Demand coordination & sequencing	✓	Fabrication facilities serve multiple sectors; coordinated demand signals may help manage production queues
R&D for alternative materials & processes	✓	Advances in welding automation and fabrication processes could improve throughput and reduce production constraints

Figure 22: Box 10.3.1. Finned Tube Mitigation Pathway Screening

The industry already addresses elements of these challenges. Domestic fabrication facilities continue to operate in the U.S, while steel and tube producers have announced selective investments in higher-value alloy processing. HRSG production remains closely tied to global OEM supply chains and project-driven demand cycles (See Box 10.3.2).

10.3.2. Finned Tubes: Current Mitigation Landscape		
Relevant Mitigation Levers	Existing Actions	Remaining Gaps
Domestic capacity expansion	• Domestic consolidation and fabrication capability e.g., B&W Chanute facility activity to compete with large international HRSG suppliers (<i>Babcock & Wilcox 2022</i>) • Selective U.S. steel mill investments targeting higher-value alloy and finishing capabilities (<i>Tenaris 2026; Vallourec 2025; U.S. Steel 2025</i>)	• Domestic capacity not configured for large-scale new-build HRSG demand • Expansion constrained by high capital costs for welding lines and heat-treatment furnaces • Business case uncertain due to project-driven HRSG demand volatility
Allied sourcing	• Completed finned-tube harps sourced primarily from allied suppliers in East Asia e.g., Vietnam, South Korea, Japan (<i>McCoy Power Reports 2024</i>) • Limited PRC exposure for finished HRSG assemblies	• Supplier base highly concentrated among vertically integrated OEM fabrication networks • Foreign production locations may limit U.S. buyers' influence over production schedules, particularly during demand surges
Demand coordination & sequencing	Informal market-driven alignment emerging from growth in AI data-center electricity demand, nuclear deployment, and gas-turbine expansion (<i>Babcock & Wilcox 2025; Kelvion 2025; Clean Air Task Force 2025; American Iron and Steel Institute 2025</i>)	• No formal cross-sector demand coordination mechanism • Year-to-year volatility in HRSG orders continues to deter domestic finning-line investment
R&D / process improvements	Adoption of advanced welding technologies e.g., laser welding beginning in finned-tube fabrication (<i>Del Fine Tubes 2023</i>)	R&D may improve competitiveness but does not resolve near-term capacity or scale constraints without sustained demand

Figure 23: Box 10.3.2. Finned Tube Current Mitigation Landscape

Building on the mitigation landscape discussed above, Box 10.3.3 summarizes priority pathways. Mitigation actions could include stabilizing HRSG demand signals, maintaining access to allied fabrication capacity, and selectively supporting domestic fabrication expansion where credible long-term demand exists.

Box 10.3.3. Finned Tubes: Prioritized Mitigation Pathways	
Relative Impact	Mitigation Options & Illustrative Examples
Near-term impact	Stabilize HRSG demand signals to support domestic finned-tube investment: Facilitate coordination of long-term demand visibility across gas, nuclear, and data-center power development to reduce order volatility and support domestic fabrication investment Secure access to allied fabrication capacity: Strengthen sourcing relationships with HRSG and finned-tube suppliers in allied Asia-Pacific production hubs (Vietnam, South Korea, Japan) through targeted trade and financing tools, in partnership with OEMs, to ensure U.S. buyers retain priority access during demand surges
Mid-term impact	De-risk targeted domestic fabrication capacity expansion: Engage industry to incentives tied to credible power-sector demand for investments in finning lines, welding systems, and heat-treatment furnaces Strengthen domestic specialty steel tubes supply (e.g., T91): Use DOE grants and loan programs to support production of high-temperature steels and alloy tubes used in HRSG pressure parts and heat-exchange components
Longer-term impact	Advance R&D in welding process innovation: Support demonstration of advanced welding techniques (e.g., laser welding) to improve productivity and competitiveness of domestic finned-tube manufacturers
Note: Relative impact reflects a qualitative assessment of when each mitigation option is likely to begin meaningfully influencing the identified supply-chain constraint, considering practical implementation timelines	

Figure 24: Box 10.3.3. Finned Tube Priority Mitigation Actions

11 Conclusion

This study examined supply-chain exposure across major power-plant subsystems to identify supply chain constraints that could affect power-generation deployment timelines. The system-level screening confirms that supply-chain risk is present across the power-plant architecture but is concentrated in a limited number of material inputs and specialized manufacturing stages.

Across the eleven subsystems evaluated, thirteen components were identified where material dependencies or manufacturing capacity constraints could create potential deployment challenges. These components include upstream alloy inputs, specialized turbine manufacturing processes, heavy forging infrastructure, and heat-exchange fabrication systems. While many parts of the power-plant supply chain rely on mature industrial production networks, these areas represent supply chain stages where disruptions could spread across multiple subsystems and influence plant construction schedules.

To examine these risks in greater detail, the analysis focused on five areas selected to align with the scope and timeline of the sprint: nickel and chromium, turbine blades and vanes, large forgings used in turbine rotors, and finned tubes used in heat-recovery steam generators. The analysis highlights how upstream material availability, specialized capital equipment, and manufacturing throughput are interconnected and influence supply-chain security for power-generation infrastructure.

The findings indicate that strengthening power-plant supply chains will depend less on broad system-wide interventions and more on targeted efforts at key upstream materials and manufacturing stages. Improving throughput at existing facilities, reinforcing supply-chain diversification through allied partnerships, and supporting investment in specialized manufacturing capacity are likely to have the greatest influence on near-term deployment timelines.

Future work can expand this analysis to additional subsystems and components identified during the screening process. Several areas, including electrical systems, control equipment, advanced materials, and capital equipment supply chains, showed elevated risk signals but were not examined in detail due to sprint scope constraints. Continued analysis in these areas will help refine understanding of system-level vulnerabilities and inform strategies to strengthen industrial capacity and support reliable power-generation deployment.

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A Supply Chain Risk Assessment Methodology

Appendix A describes the analytical framework used to evaluate supply-chain risk across power-plant subsystems and support the detailed screening of the six prioritized subsystems examined in this report. The methodology evaluates supply-chain exposure using five criteria that capture key structural sources of risk: cross-technology use, foreign input dependence, domestic manufacturing capacity, critical mineral dependency, and manufacturing lead times. These criteria were selected intentionally from many possible indicators because they reflect factors that strongly influence whether the United States can manufacture or reliably access key power-plant components when needed.

The analysis was conducted using a two-stage screening framework. The first stage applied an initial subsystem screening methodology to identify areas with potential supply-chain exposure across power-plant subsystems. The second stage applied a detailed risk-screening framework to the six prioritized subsystems to evaluate component-level supply-chain risks.

Table A-1 presents the initial subsystem screening methodology used to flag supply-chain exposure across subsystem component families.

Table A-2 presents the detailed subsystem risk-screening framework used to evaluate component-level supply-chain risk within the six prioritized subsystems. Each criterion is assessed using a three-tier classification system representing low, moderate, and high supply-chain risk, based on available industry data, public reports, and analyst assessment of supply-chain conditions.

Table A-1: Initial Subsystem Screening Methodology

Evaluation Criterion	Rationale	Where the Logic is Applied	Assessment Logic
Cross-Technology Use	Flags shared risk exposure.	Final Assembly	1 point if used across 2 power plant types; 2 points if used across 3.
Input Foreign Dependence	Flags exposure to foreign supply of inputs.	Processing & Refining to Engineered Materials	1 point per flagged material (Processed/Refined + Engineered) sourced from high-risk or single-country supply (e.g., PRC).
Domestic Manufacturing Capacity	Evaluates resilience of U.S. manufacturing and OEM presence.	Subcomponents to Final Assembly	1 point if limited U.S. OEM production; 2 if no capacity
Critical Material Dependency	Flags reliance on strategic minerals.	Processed & Refined	1 point per flagged critical material (Ni, REEs, Co); add +1 bonus if multiple appear in same component.
Lead Time	Flags manufacturability and deployment delays.	Subcomponents + Component Assemblies	1 point per flagged part in Subcomponent + Component Assemblies with long lead times or forging constraints.

Table A-2: Detailed Subsystem Risk Screening Framework

Evaluation Criteria	Description	Low Risk	Moderate Risk	High Risk
Cross-Technology Use	Degree to which the product is used across NGCC and other power generation technologies (geothermal, hydropower, other renewables)	Used only in NGCC + other sectors/technologies	Used in NGCC + 1–2 other power plant types + other sectors/technologies	Used in NGCC + 3 or more power plant types and other sectors/technologies
Input Foreign Dependence	Share of foreign-sourced inputs (domestic capacity of the prior stage of the supply chain e.g., Bauxite for Aluminum or Polysilicon for Wafers)	0 – 10%	10 – 50%	> 50%
Domestic Manufacturing Capacity	Availability of U.S. OEMs / qualified suppliers	Multiple U.S. suppliers (range is industry-dependent)	Limited U.S. suppliers (range is industry-dependent)	No meaningful U.S. production (range is industry-dependent)
Critical Mineral Dependency	Degree to which product requires CMs (Ni, Co, REEs, etc.)	No Critical Mineral use or is not CM (upstream)	Uses CMs with manageable but concentrated supply chains or is a CM with manageable concentration (upstream)	Use multiple Critical minerals or is a CM that is highly concentrated in China (upstream)
Lead Time	Manufacturing cycle duration and flag complexity	Less than 10 weeks	10 – 20 weeks	Over 20 weeks

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